

EENG30013 POWER ELECTRONICS MACHINES AND DRIVE TECHNOLOGIES DESIGN

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Contents

EENG30013 POWER ELECTRONICS MACHINES AND DRIVE TECHNOLOGIES DESIGN	1
Introduction	3
Machine	3
1. Part.A	4
1.1 Elevator's acceleration and torque of drum	4
1.2 Output and input angle velocity and acceleration	4
1.3 Damping torque and Torque of mass moment of inertia	5
1.4 Power transmission efficiency	5
1.5 Result of data analyze in Part.A:	5
1.6 Angular Velocity Input (Blue section):	5
1.7 Torque Calculation:	6
1.8 Other scope's Outputs:	6
1.9 Result of Simulation model in Part.A	6
1.10 Profound consideration of Part.A	6
2. Part.B	7
2.1 Current calculation for P1 and P2	7
2.2 Voltage calculated for P1 and P2	7
3. Part.C	8
3.1 Voltage input	8
3.2 Simulate Voltage input	8
3.3 Simulation result of three-phase voltage and current	9
3.4 Simulation Result of V_{dq0} and i_{dq} under tow operating points	9
3.5 Complete torque diagram	10
3.6 Efficiency of the electro-mechanical energy conversion	11
3.7 Summary and Profound consideration of Machine.Part	11
4. Part.D Converter	11
4.1 Capacitor selection	12
4.2 Capacitor current	12
4.3 Inductance and Resistance	13
4.4 DC Voltage Ripple	13
4.5 Capacitor voltage ripple	14
4.6 Harmonic spectrum for converter(P1 and P2)	14
5. Part.E	15
5.1 Power loss	15
5.2 Summary of Part.D and Part.E	15
6. Part.F POWER ELECTRONICS (SWITCHES)	15
6.1 Harmonic spectrum for Switch(P1 and P2)	16
6.2 Power loss	17
7. Part.G Heat sink	18
8. Part.H Summary for all elevator system	19

Introduction

The aim of this report is to analyze and design an electric drive system using the high-speed elevator of the ‘Shard Building’ in London as a virtual case investigate. This project examines an electric drive system that uses a conventional 3-phase, 2-level inverter to drive a three-phase AC permanent magnet synchronous motor (PMSM). The motor drives an express elevator by delivering torque through a gearbox connected to a permanent magnet synchronous motor via a shaft. The gearbox's output connects to a drum around which lifting wires are wound and attached to the lift. The lift has a counterweight which matches 100% of the lift carriage mass (600 kg) plus 40% of the design load (10 people, each 80 kg). The DC input rectified to the inverter is 325V. This report will detail the components of the system from MACHINE, DRIVE (CONVERTER) and POWER ELECTRONICS (SWITCHES) with its principle of operation, and its ability to achieve efficient and stable operation during elevator driving. The complete structure of the drive system is shown in Figure.1 and will serve as the basis for subsequent analysis and simulation.

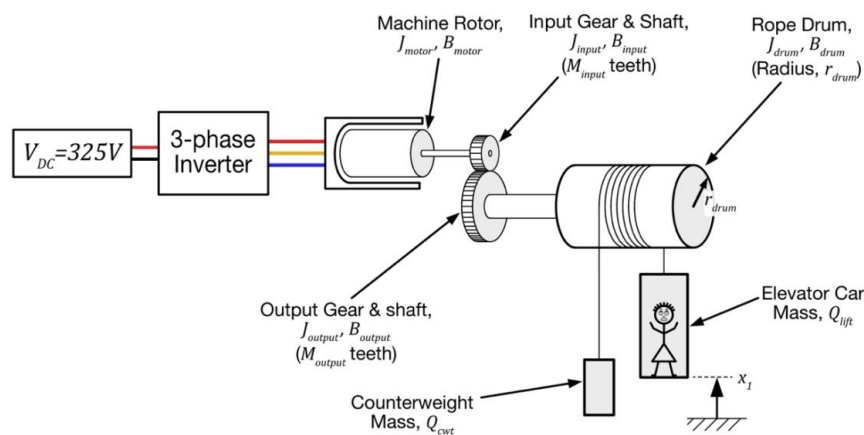


Figure.1: Elevator Drivetrain[1]

Machine

The mechanical part of the this project has three objectives.

- A. Transform the given speed Vs time profile into a motor torque Vs time curve and prove its correctness by simulating it on MATLAB.
- B. Using the machine parameters to calculate the voltages, currents, and losses at the two given operating points P1($t = 3.49s$) and P2($t = 88.25s$) on the speed vs time profile shown in Figure 2.
- C. Simulate the operation in MATLAB and validate the correctness of calculated voltages and currents, and assess the efficiency of the electro-mechanical energy conversion process at these points.

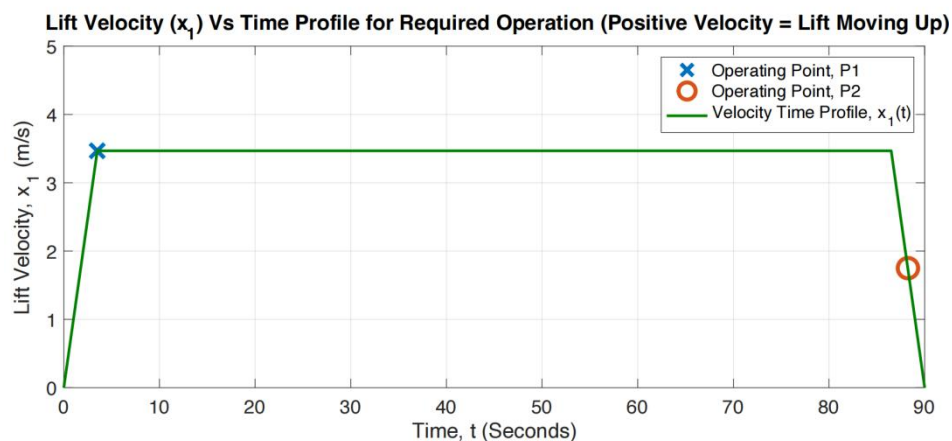


Figure.2: Required linear velocity Vs Time profile[1]

1. Part.A

Methods for validation: To achieve the aim of Part A, This report use two methods for validation: Data analysis and Simulation modeling. Data analysis provides theoretical motor torque vs time equations under ideal conditions, Simulation modeling provides a reflection of real-world performance.

Data analyzing

1.1 Elevator's acceleration and torque of drum

To calculate the torque vs time profile, the acceleration of the elevator was first determined up to its, Based on the linear velocity Vs Time profile in Figure.1 and the acceleration formula(1) where ΔV is the changing velocity and Δt is changing time

$$\alpha = \frac{\Delta V}{\Delta t} \quad (1)$$

$$F = ma \quad (2)$$

the elevator's acceleration can be determined shown in table.1. In this report, the gravitational acceleration g is set to 9.81 m/s^2 and the radius(r) of drum set as 0.2m , the gravitational force(F) on both sides of the elevator can be calculated using the formula(2) where m is the mass.

$$\Delta F = [m_{\text{life}} \times (g + a)] - [m_{\text{cwt}} \times (g - a)] \quad (3)$$

$$T = \Delta F \times r \quad (4)$$

Through formula(3) and(4), drum torque can be calculated shown in table.1.

Type of drum motion	Running Time	Elevator's acceleration	Torque of drum
acceleration motion	0 - 3.5s	1 m/s^2	$T_{\text{drum}} = 1405.76\text{Nm}$
constant velocity motion	3.5–86.5s	0 m/s^2	$T_{\text{drum}} = 951.77\text{Nm}$
deceleration motion	86.5 - 90s	-1 m/s^2	$T_{\text{drum}} = 477.76\text{Nm}$

Table.1 :Data of Elevator's acceleration and torque of drum

1.2 Output and input angle velocity and acceleration

In analyzing the drum's motion, Through formula(5) and(6) the output angle velocity(ω_{out}) can be calculated and shown in table.2. In this analyzing, three scenarios has been considered; drum at acceleration, constant velocity and deceleration motion.

$$\omega(t) = \frac{\alpha}{r} \times t \quad (5)$$

$$\omega_{\text{out}} = \frac{V}{r} \quad (6)$$

To calculate the input angular velocity(ω_{in}), the gear ratio(m) must first be determined. Using formula (7), where M_{in} is the number of teeth on the input gear and M_{out} is the number of teeth on the output gear, m can be calculated as $m = 3$. Therefore, the input angular velocity can be calculated using formula (8).

$$m = \frac{M_{\text{out}}}{M_{\text{in}}} \quad (7)$$

$$\omega_{\text{in}} = m \times \omega_{\text{out}} \quad (8)$$

$$\alpha = \frac{d\omega}{dt} \quad (9)$$

As same as the output angle velocity, input angle velocity(ω_{in}) also consider in three scenarios. At the same time according to formula (9), the angular acceleration can also be obtained directly which shown together with the angle velocity in table.2 .

Type of drum motion	Input angle velocity	Output angle velocity	Input angle acceleration	Output angle acceleration
acceleration	$15t \text{ rad/s}^2$	$5t \text{ rad/s}^2$	15rad/s^2	5rad/s^2
constant velocity	52.5rad/s^2	17.5rad/s^2	0 rad/s^2	0 rad/s^2

- Acceleration Phase (0–3.5s): The angular velocity linearly increases to simulate the acceleration of the elevator.
- Constant Velocity Phase (3.5–86.5s): The angular velocity remains constant to maintain the elevator's steady motion.
- Deceleration Phase (86.5–90s): The angular velocity decreases linearly to simulate deceleration.

1.7 Torque Calculation:

- Using ideal torque sensor to measures the motor's torque in real-time based on input angular velocity and system dynamics.
- Direct indication the relationship between motor torque and time

1.8 Other scope's Outputs:

- Input Angular Velocity and Acceleration which shown in Figure.4
- Output Angular Velocity and Acceleration which shown in Figure.5

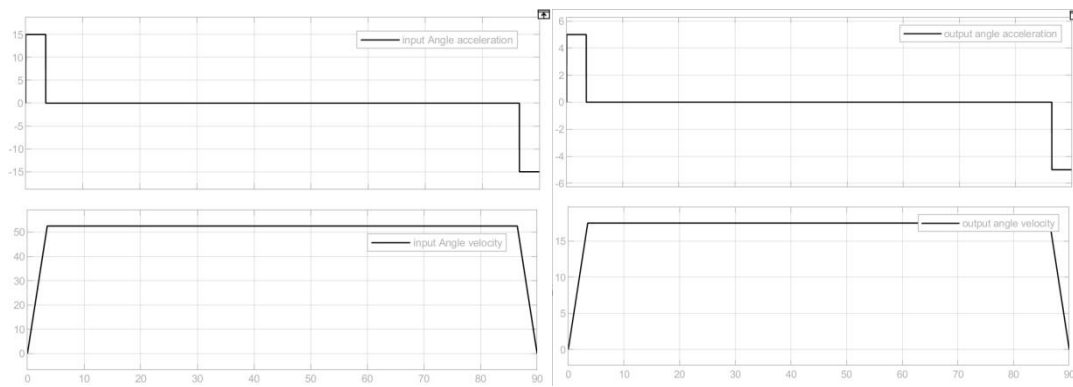


Figure.4: data of input a and ω

Figure.5: Data of output a and ω

1.9 Result of Simulation model in Part.A

Graph of the motor torque vs time which detected by the torque sensor was shown in Figure.6. The torque vs time graph demonstrates the expected system behavior, It has a peak torque during acceleration, stable torque during constant velocity, and a rapid decrease during deceleration.

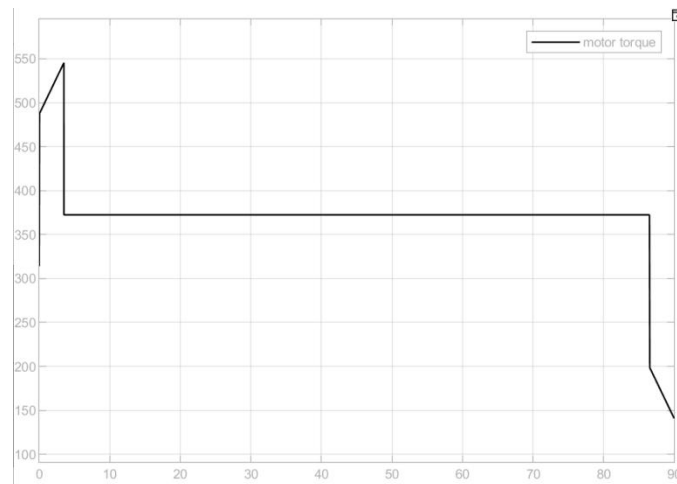


Figure.6: Torque vs time profile

1.10 Profound consideration of Part.A

Comparison of the simulated torque versus time curves with the three torque versus time equations derived from the data analysis shows that they are in almost perfect agreement. The consistent results are mainly due to two key factors:

- First, this simulations were performed in an ideal environment and does not consider potential factors such as mechanical overheating, machine component wear, or electrical losses within the motor. In real-world applications, these factors may slightly alter torque and efficiency.
- Second, A 100% power transmission efficiency between input and output has been assumed.

2. Part.B

In Part B, the results of Part A are first utilized to determine the torque of the motor at two operating points (P1: $t = 3.49s$, P2: $t = 88.25s$). Which P1 represent the peak motor torque at maximum positive speed and P2 is the midpoint of the deceleration phase, respectively. From the simulation results shown in Figure.7, it can be concluded that the torque values at these two points are 545.222 Nm (P1) and 170.121 Nm (P2). These motor torque values will be used as a basis for further calculations of the corresponding voltages, currents and system losses.

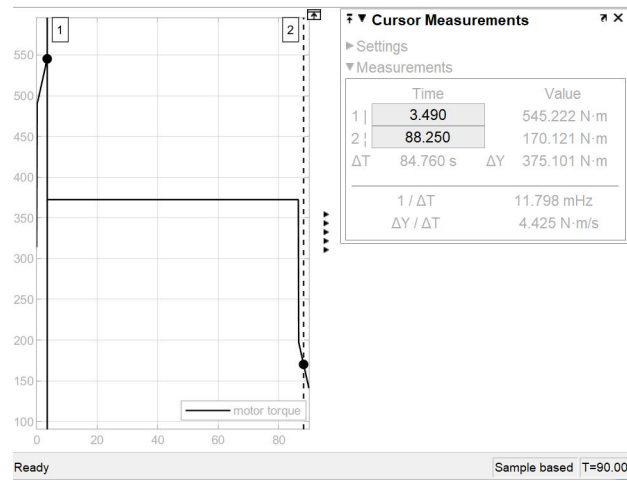


Figure.7: Motor Torque at P1 and P2

2.1 Current calculation for P1 and P2

$$T_{em} = \frac{3}{2} P (i_{qs} \lambda_{PM} + i_{ds} i_{qs} (L_{ds} - L_{qs})) \quad (12)$$

Through formula(12) , the current at both point P1 and P2 can be calculated. Since the permanent magnet provides a fixed flux λ_{PM} (Permanent Magnet Flux) and $L_{ds} = L_{qs} = 0.00285$, this indicates that the flux in the d-axis (magnetic flux direction) is entirely provided by the permanent magnet, so there is no need for excitation current i_{ds} to contribute additional flux. Therefore, i_{ds} is always maintained at 0. So the value of i_{qs} could be direct calculated shown in table.5

Operating points	Time	i_{ds}	i_{qs}
P1	3.49s	0A	140.76A
P2	88.25s	0A	43.92A

Table.5:Data for current at two point

2.2 Voltage calculated for P1 and P2

$$\omega_e = P \omega_{in} \quad (13)$$

$$v_{qs} = i_{qs} R_s + L_{qs} \frac{di_{qs}}{dt} + \omega_e (\lambda_{PM} + L_{ds} i_{ds}) \quad (14)$$

$$v_{ds} = i_{ds} R_s + L_{ds} \frac{di_{ds}}{dt} - \omega_e L_{qs} i_{qs} \quad (15)$$

From formula(13),the angle velocity at point P1 and P2 can be derived, and through formula(13) ,(14) , the Voltage at both point P1 and P2 can be calculated shown in table.6

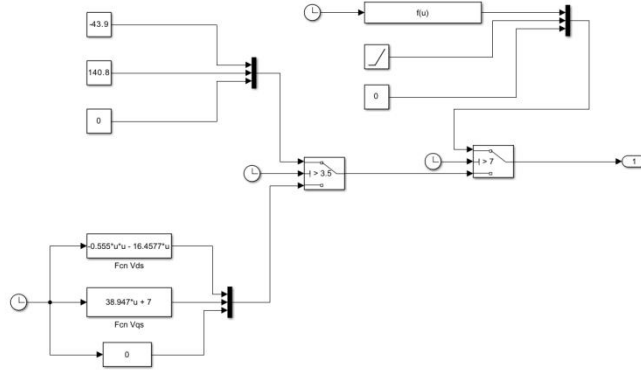


Figure.9: Simulate Voltage input

This change makes the simulation faster and helps analyze voltage, current, and torque during the three main phases.

3.3 Simulation result of three-phase voltage and current

Figure.10 shows the simulation result for all periods

1. During the acceleration phase (0-3.5s), Both the three-phase currents and voltages increase during this period.
2. During the constant velocity phase(3.5-7s), Both three-phase voltage and current keep stable
3. During the deceleration phase (7-10.5s), Both the three-phase currents and voltages decrease during this period.

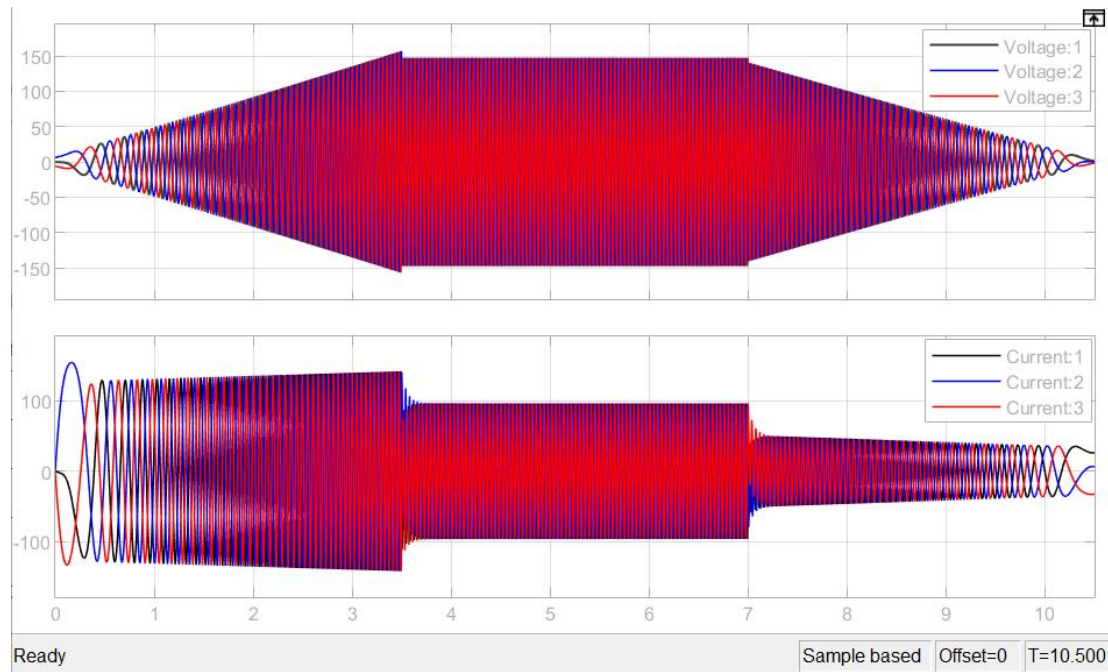


Figure.10: Result for voltage and current(0-10.5s)

3.4 Simulation Result of V_{dq0} and i_{dq} under tow operating points

For the acceleration part(0-3.5s) and deceleration(7-10.5s):

- The result of V_{qs} and V_{ds} at point P1 and P2 was shown in Figure.11 and.13, which Park 1:2 is V_{qs} and Park 1:1 is V_{ds} ;
- The result of i_{qs} and i_{ds} at point P1 and P2 was shown in Figure.12 and.14, which Park.2 is i_{qs} and Park.1 is i_{ds} .

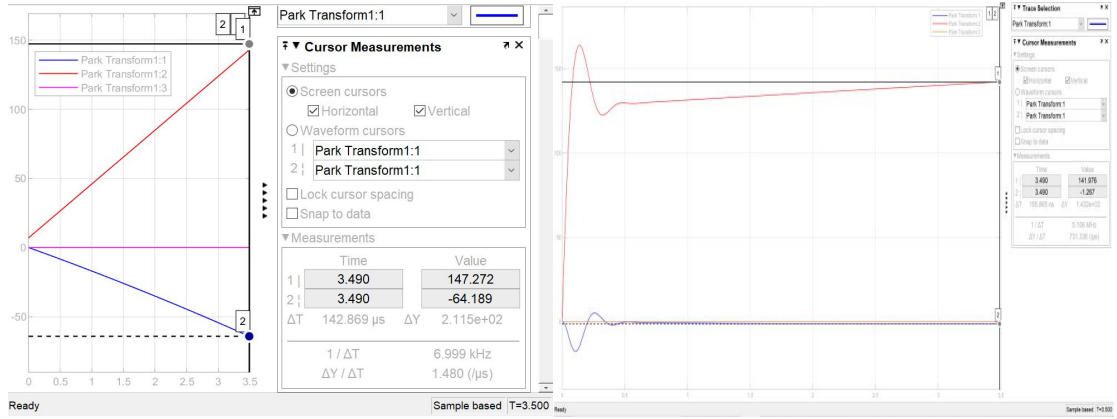


Figure.11: Value of V_{qs} and $V_{ds}(Acc)$

Figure.12: Value of i_{qs} and $i_{ds}(Acc)$

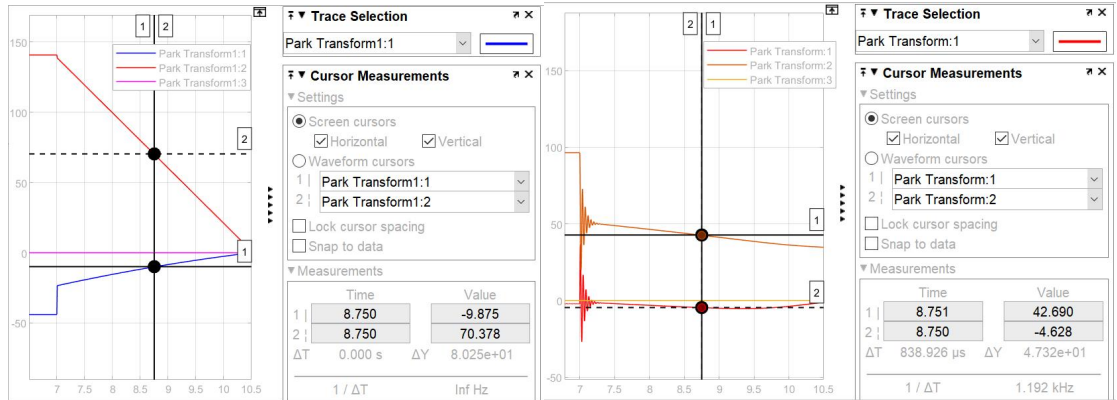


Figure.13: Value of V_{qs} and $V_{ds}(Dec)$

Figure.14: Value of i_{qs} and $i_{ds}(Dec)$

All the result was shown in table.8

Operating points	V_{qs}	V_{ds}	i_{qs}	i_{ds}
P1(3.49s)	147.272V	-64.189V	142A	-1.26A
P2(88.25s)	70.378V	-9.875V	42.69A	-4.628A

Table.8:Data for V_{dq0} and i_{dq}

3.5 Complete torque diagram

The motor torque vs time profile under all machine components is shown below in Figure.15. And compare the motor torque results in Figure.15 and Figure.7, they are very close that demonstrating the feasibility of the mechanical part.

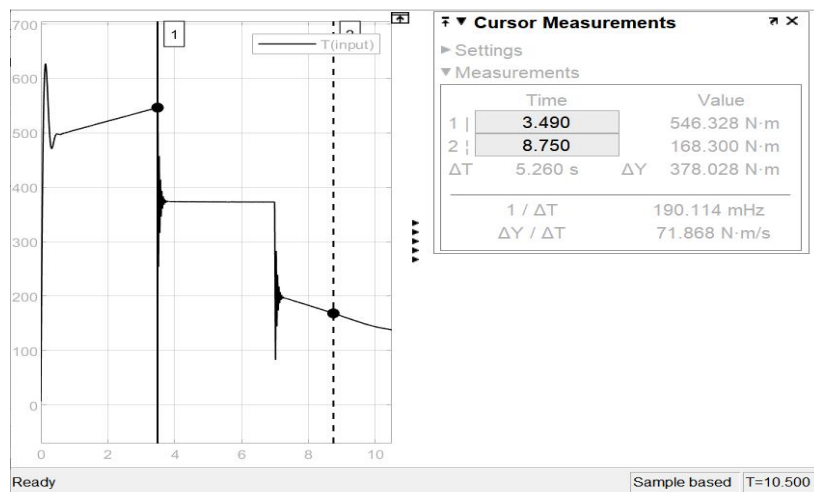


Figure.15: Result of motor torque vs time profile

3.6 Efficiency of the electro-mechanical energy conversion

Through the formula(18),(19) and (20), the final power conversion efficiency could be calculated, shown in table.9

$$P_{in} = \frac{3}{2}(V_{ds}i_{ds} + V_{qs}i_{qs}) \quad (18)$$

$$P_{out} = T_{em}\omega_{in} \text{ and } P_{drum} = T_{drum}\omega_{out} \quad (19)$$

$$\eta_{mach} = \frac{P_{out}}{P_{in}} \text{ and } \eta_{total} = \frac{P_{drum}}{P_{in}} \quad (20)$$

Operating points	P_{in}	P_{out}	P_{drum}	η_{mach}	η_{total}
P1(3.49s)	30125W	28540W	24530W	95%	81%
P2(88.25s)	4620W	4463W	4180W	96.6%	90%

Table.9:Data for power and conversion efficiency

3.7 Summary and Profound consideration of Machine.Part

- In analyzing the mechanical part, theoretical data analysis and MATLAB simulation were used to achieve the goal. Combining theory and simulation gave a full analysis of motor behavior. The results showed that the mechanical design meets the elevator system's requirements.
- The simulation output graph Figure.10 shows large oscillations at the start of each phase because the motor must reach the maximum current needed for acceleration or deceleration. In reality, this increase is more gradual. This is because the simulation voltage input is no connected basically, which causes big current ripples when each state changes. Over time, the current stabilizes and gets closer to the theoretical values.
- MATLAB simulation gave a more realistic view of system performance. It included factors like inertia, damping torque, and sudden phase transitions. But the simulation time had to be reduced, such as by shortening the constant speed phase to match computing limits. This may affect the accuracy of simulation.
- However, MATLAB only operates on a two-dimensional platform. Its simulations and calculations are limited to analyses under ideal input conditions. This can be restrictive when dealing with real-world three-dimensional dynamics and nonlinear behaviors.

4. Part.D Converter

The Simulation model Driver was shown in Figure.15,This simulation model uses a 325-volt DC source as input, converted to AC by a three-phase inverter. The model includes a PWM generator to control inverter switch frequency and phase. And some of background data has been selected shown in table.10

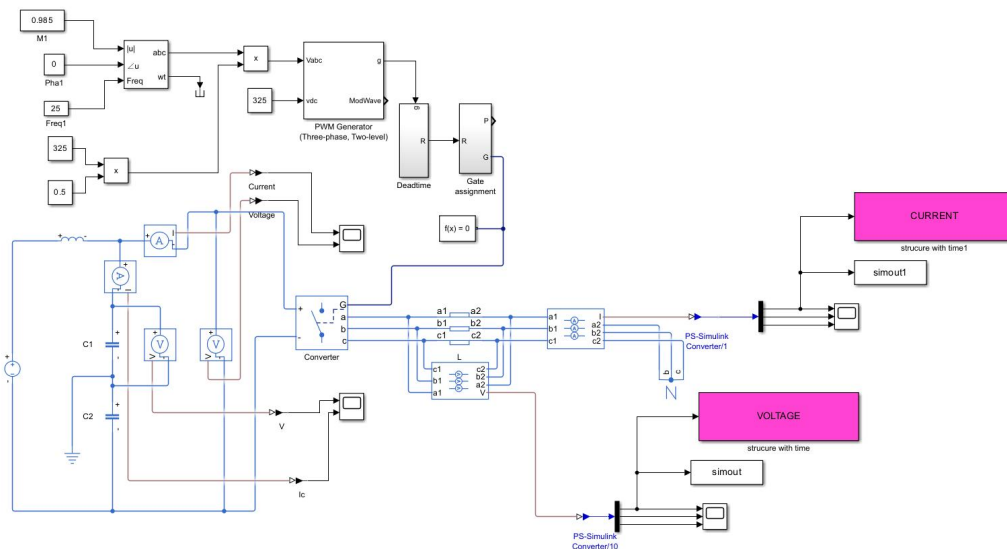


Figure.15: Simulation model of converter

Background data	Value
T_s	5e-7s
f_{sw}	2500Hz
ΔV_{ripple}	5%
Δi_{ripple}	5%
V_{DC}	325V

Table.10:Setting data in part of driver

4.1 Capacitor selection

For the analysis the converter at point P1 and P2, first to calculate the modulation index M, through formula(21) and (22) the phase voltage, current and modulation index can be calculated as shown in table.11

$$V_{phase} = \sqrt{V_{qs}^2 + V_{ds}^2} \text{ and } i_{phase} = \sqrt{i_{qs}^2 + i_{ds}^2} \quad (21)$$

$$M = m_i = \frac{V_{phase}}{V_{DC/2}} \quad (22)$$

$$C_{DC_P1}(M = 0.5 - 1) = \frac{i_{phPK}}{8f_{sw}\Delta V_{ripple}} \text{ and } C_{DC_P2}(M = 0.1 - 0.5) = \frac{i_{phPK}}{4f_{sw}\Delta V_{ripple}} \quad (23)$$

The result of the capacitor shown in table.11 calculation from Formula(23) is used to help select the appropriate capacitor to ensure the voltage ripple on the DC link is kept within an acceptable range. Which can optimize the performance of the inverter and reduces the potential for equipment damage due to voltage fluctuations.

Operating points	V_{phase}	i_{phase}	Modulation index	C_{DC}	Capacitor Choose
P1(3.49s)	160V	142A	0.985	874uF	4700uF
P2(88.25s)	71.1V	43A	0.438	529uF	4700uF

Table.11:Data for phase voltage and current, modulation index and Capacitor

4.2 Capacitor current

To calculate current through the capacitor, base on the output results in Figure.16 which from Machine part, the load angle between three-phase current and voltage at two operation points can be measured.

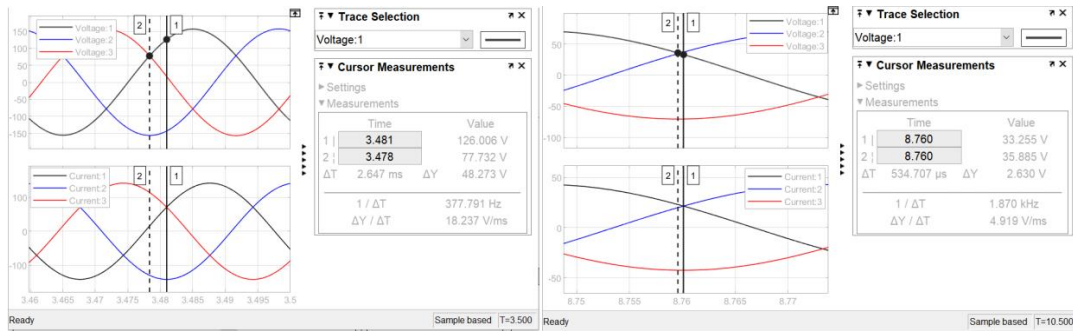


Figure.16: Load angle between current and voltage

$$i_{RMS} = \frac{i_{peak}}{\sqrt{2}} \text{ and } \theta = 2\pi f\Delta t \quad (24)$$

$$i_{cRMS} = i_{phRMS} \sqrt{2M \left[\frac{2\sqrt{3}}{\pi} + \cos^2(\theta) \left(\frac{\sqrt{3}}{\pi} - \frac{9}{16}M \right) \right]} \quad (25)$$

Through formula(24) and(25),the data of Capacitor current can be derived shown in table.12

Operating points	i_{phRMS}	θ	i_{cRMS}	i_c
P1(3.49s)	100.4A	0.416rad	51.88A	73.4A

P2(88.25s)	30.4A	0.042rad	33.75A	47.7A
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Table.12:Data for phase angle and RMS current

4.3 Inductance and Resistance

The total impedance Z can be direct calculated with the RMS value of phase voltage and current, and it can also expressed in terms of its real part (resistance R) and its imaginary part (inductive reactance X_L) shown in formula(26). The magnitude of the total impedance represents the combination of resistance and inductive reactance. This allows for the calculation of the actual values of resistance R and inductance L in the load.

$$X_L = \omega L \text{ and } |Z| = \frac{V_{phRMS}}{I_{phRMS}} = \sqrt{R^2 + (\omega L)^2} \quad (26)$$

$$R = |Z| \cos \theta \quad (27)$$

Additionally, the effect of resistance R can be directly determined by multiplying the phase angle with the total impedance show in formula(27). Through the formula(26) and(27) the value of Inductance and Resistance at point P1 and P2 can be confirmed shown in table.13

Operating points	Z	L	R
P1(3.49s)	1.12	2.77mH	1.03Ω
P2(88.25s)	1.653	1.03mH	1.651Ω

Table.13:Result of Z, R and L

After determining the values of resistance and inductance, the RMS current of the capacitor at points P1 and P2 was measured and recorded shown in Figure 17,which RMS currents was 56A at P1 and 19.5A at P2.



Figure.17:Result of capacitor RMS current at point P1 and P2

4.4 DC Voltage Ripple

The voltage ripple setting as 5% of the DC voltage (325 V) is demonstrated through simulation at points P1 and P2. The measured voltage fluctuations at these points are shown Figure.18..

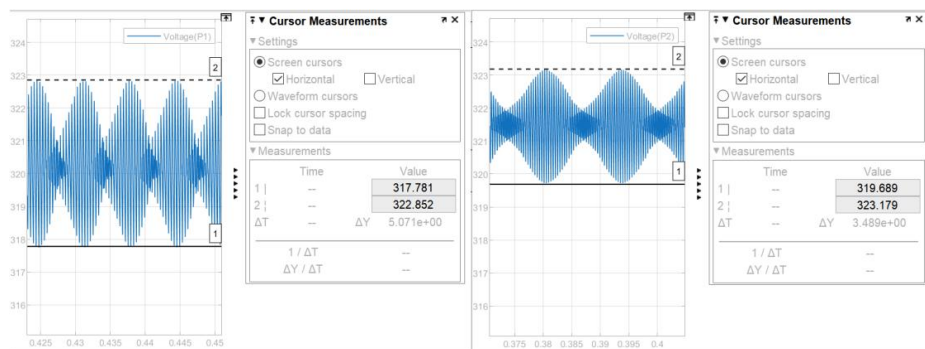


Figure.18:Result of Voltage ripple at point P1 and P2

The voltage ripple at P1 is 5.071V (1.5%), and at P2 it is 3.49V (1%), both complying the expected standard for voltage ripple setting under varying electrical loads

4.5 Capacitor voltage ripple

The result of capacitor voltage in point P1 and P2 was shown in Figure.18.5; The capacitor voltage ripple at P1 is 2.535V (1.5%), and at P2 it is 0.61V (0.4%). Both of the voltage ripples are much lower than the setting range(5%), Compliance with the system load standards.

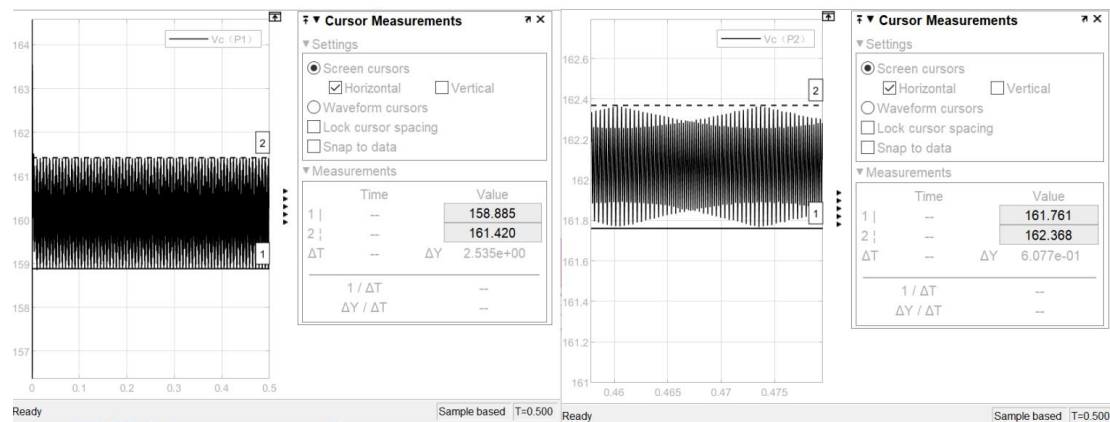


Figure.18.5:Result of capacitor Voltage ripple at point P1 and P2

4.6 Harmonic spectrum for converter(P1 and P2)

Total Harmonic Distortion (THD) is a measure of signal or power quality. It shows the proportion of all harmonic frequencies to the fundamental frequency. Lower THD suggests better signal quality and stable operation of electrical equipment, while high THD can lead to heating and reduced efficiency. Use the FFT tool, the current and voltage spectrum analysis of operation point P1 was shown in Figure.19 and the data of spectrum was shown in table.14

Spectrum	Fundamental(25Hz)	THD
Current	140.1	1.24%
Voltage	156.7	66.52%

Table.14:Data of current and voltage spectrum of point P1

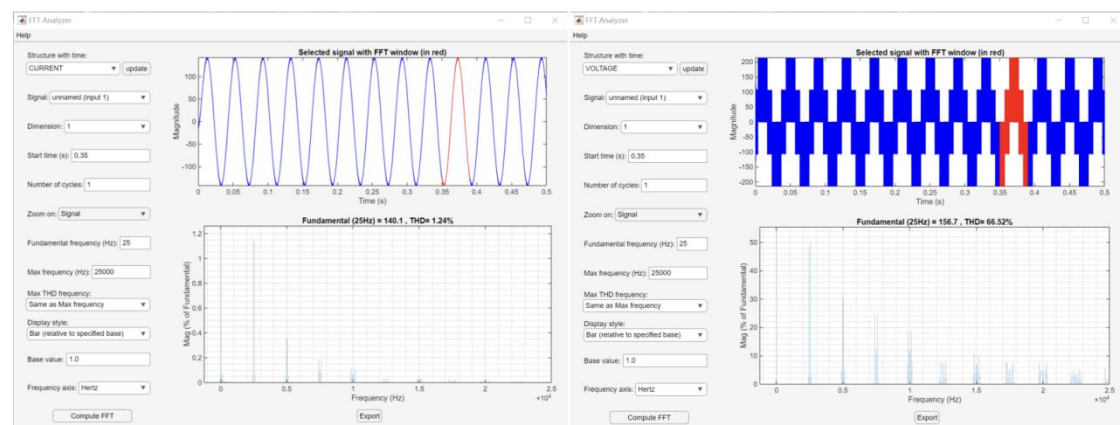


Figure.19:Current and voltage spectrum for point P1(Converter)

The current and voltage spectrum analysis of operation point P2 was shown in Figure.20, and the data of spectrum was shown in table.15

Spectrum	Fundamental(12.5Hz)	THD
Current	42.4	6.58%
Voltage	70.08	136.36%

Table.15:Data of current and voltage spectrum of point P2

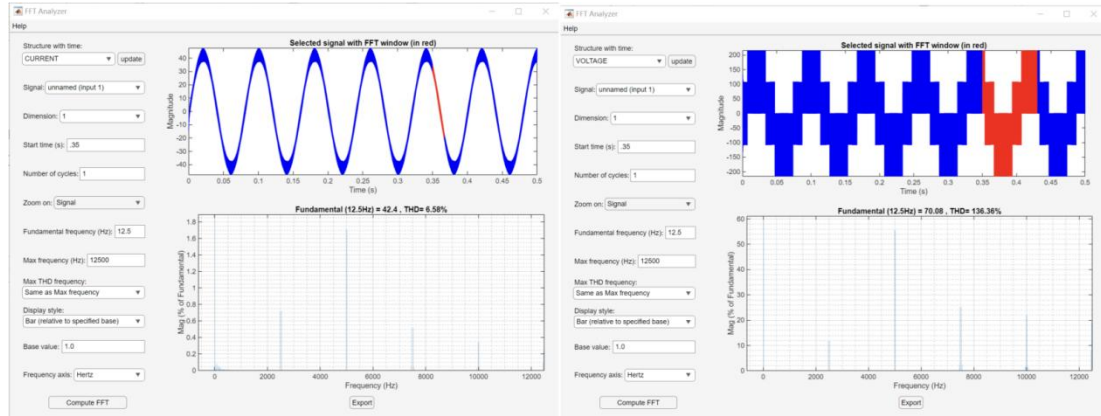


Figure.20:Current and voltage spectrum for point P2(Converter)

5. Part.E

5.1 Power loss

According to the results of the spectral analysis in the part.D, use the FFT tool, all harmonics voltage, current and phase angle was given, put all the data into Excel table shown in figure.21 and.22; through the formula(28), the power loss can be measured.

$$P = IV \cos(\theta) \quad (28)$$

#	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	frequency	number	CURRENT	current angle phase			frequency	number	VOLTAGE	voltage angle phase		angle phase	power loss	stator loss
2	0	DC'	0.000769268	90			0	DC'	0.0017567710763538	90			$P = V \cos(\theta)$	2.745280851
3	25	Fnd'	140.1249673	244.558125477940			25	Fnd'	156.67788880809	267.460260641884		0.921170904		3
4	50	h2'	0.004871296	7.91679775596585			50	h2'	0.00700553516815533	48.8918923908033		0.754994685	2.5765E-05	
5	75	h3'	0.0562484460180939	206.602009991889			75	h3'	0.0927205268289613	258.298932162613		0.619821188	0.003232606	Total loss at P1
6	100	h4'	0.001523386	55.6714184028450			100	h4'	0.00390566454278089	110.447351975939		0.576775498	3.43172E-06	8.235842554
7	125	h5'	0.0695762105635782	-69.56387495			125	h5'	0.169002901189442	-4.018070786		0.413965657	0.004867649	

Figure.21: Power loss at point P1

#	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	frequency	number	CURRENT	current angle phase			frequency	number	voltage	voltage angle phase		angle phase	Power loss	stator loss
2	0	DC'	0.000390670962451495	270			0	DC'	0.000574524095670499	270			$P = VI \cos(\theta)$	11.74315862
3	12.500000000000000	Fnd'	42.3962800863915	130.880611647836			12.500000000000000	Fnd'	70.0809587041519	133.687020025886		0.998800667		3
4	25	h2'	0.000826576793020846	46.8908497528947			25	h2'	0.00232604708832327	57.7608887546026		0.98205746	1.88816E-06	
5	37.500000000000000	h3'	0.0229999617702747	206.728958064487			37.500000000000000	h3'	0.0380370825043300	215.815630507313		0.987450569	0.000863873	total loss
6	50	h4'	0.000689863749576517	189.284586014666			50	h4'	0.00271190911170093	199.381291554011		0.984513262	1.84187E-06	35.22947585
7	62.500000000000000	h5'	0.0845068054198006	97.3580543082975			62.500000000000000	h5'	0.143979566830754	110.864746882384		0.972342646	0.011830739	
8	75	h6'	0.000822998865457555	-57.29926151			75	h6'	0.00141876619483846	-26.55660429		0.85947193	1.00356E-06	

Figure.22: Power loss at point P2

The data of total power shown in table.16

Operating points	Stator loss	Total power loss
P1(3.49s)	2.74W	8.24W
P2(88.25s)	11.74W	35.23W

Table.16:Data of loss at point P1 and P2

5.2 Summary of Part.D and Part.E

- The total harmonic distortion (THD) in the voltage section is significantly larger than that in the current section
 - This may be due nonlinear loads in the system, such as PWM generation and motors, which generate a large number of harmonics during operation due to the inherent instability of the power supply.
- The deviation of RMS value of the capacitor current from the simulation results
 - Due to the simulation environment running in the deceleration phase is not as complete as that in the acceleration phase
- Energy loss in deceleration phase is larger than that in acceleration phase

6. Part.F POWER ELECTRONICS (SWITCHES)

According to the result from the part.B and part.E, from the given data-sheet the MOSFET_7 has been selected as the key components for analog switch. The simulation model of switch was shown in Figure.23 and the background data of MOSFET_7 was shown in table.17

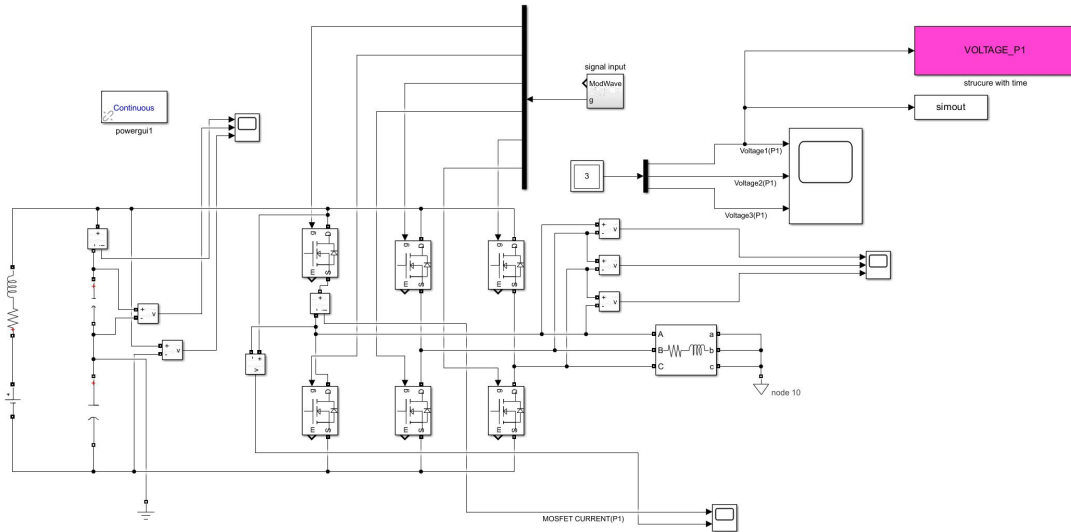


Figure.23:Simulation model for SWITCH

In this simulation model, the PWM input signal was taken directly from the part of converter(D), shown in Figure.24; By using the same PWM signal input system, and comparing the result from the Part.D at the same time, the performance of the analog switches can be analyzed more efficiently.

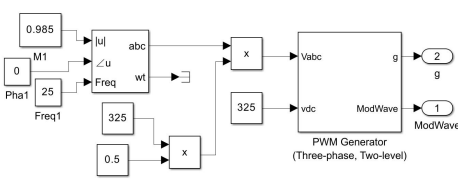


Figure.24:PWM signal input

Parameter	Value	Unit
V_{oss} over full T_j range	650	V
$R_{DS(on),typ}$	6.7	m Ω
$R_{DS(on),max}$	8.5	m Ω
Q_G,typ	179	nC
$I_D,pulse$	897	A
$Q_{loss}@400V$	337	nC
$E_{oss}@400V$	45.7	μ J

Table.17 :Key Performance Parameters of MOSFET_7

6.1 Harmonic spectrum for Switch(P1 and P2)

Through the FFT tool the harmonic spectrum of the voltage in the switch section was shown in Figure 25. The results indicate that both THD value of voltage in Converter part(Figures.19, 20) and Switch part are very close, which also confirms that the performance of MOSFET_7 is satisfy for the system requirements.The comparing data was shown in table.18

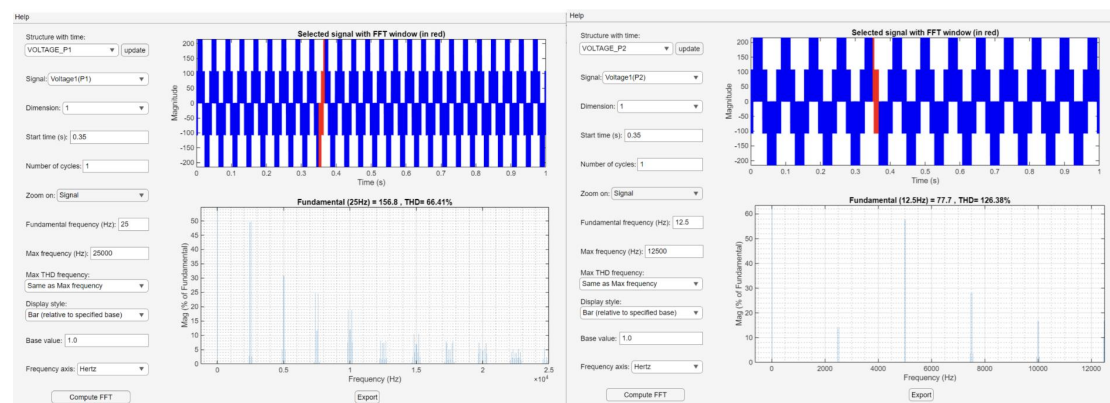


Figure.25:Voltage spectrum for point P1 and P2(Switch)

Points	Fundamental(Converter)	Fundamental(Switch)	THD(Converter)	THD(Switch)
P1(3.49s)	156.7	156.8	66.52%	66.41%
P2(88.25s)	70.08	77.7	136.6%	126.38%

Table.18:Compare data for THD and fundamental

6.2 Power loss

The energy losses in switching components primarily consist of conduction and switching losses. For calculate the conduction losses; First, measure the RMS current flowing through the MOSFET from the simulation which shown in Figure.26. Through formula(28) the conduction losses at point P1 and P2 can be derived shown in the table.19

$$P_{loss_con} = I_{mosfet_rms}^2 R_{th_on} \quad (28)$$

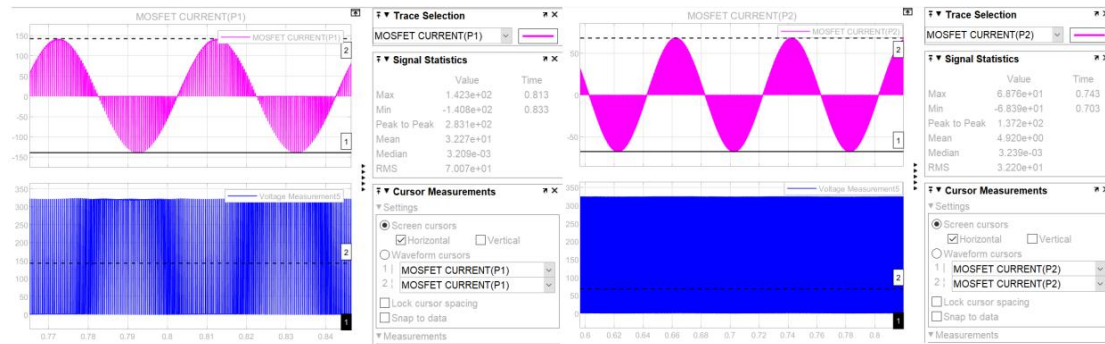


Figure.26: Voltage and RMS current of MOSFET at point P1 and P2

Operating points	I_{DS}	I_{mosfet_rms}	P_{loss_con}
P1(3.49s)	99A	70.01A	41.2W
P2(88.25s)	45.54A	32.2A	8.81W

Table.19:Data for conduction losses

Then to calculate the switching losses, using the switching losses vs current profile of MOSFET_7 shown in Figure.27(current shown in table.19), the value of E_{on} and E_{off} can be confirmed. Through formula(29), the result of switching power can be calculated shown in table.20

$$P_{loss_sw} = (E_{on} + E_{off}) \times f_{sw} \quad (29)$$

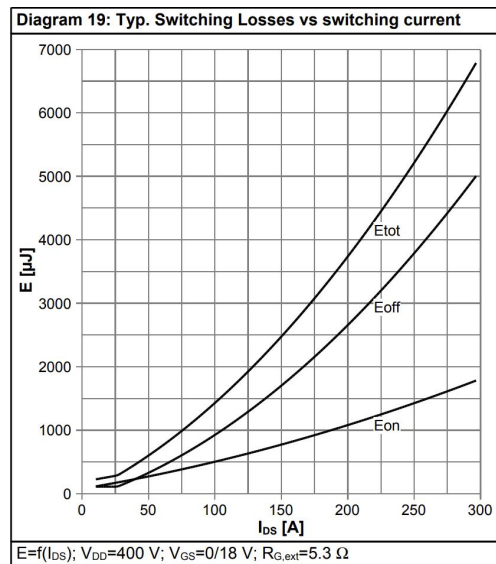


Figure.27:Switching losses vs current profile

Operating points	E_{on}	E_{off}	f_{sw}	P_{loss_sw}
P1(3.49s)	500μJ	980μJ	2500Hz	3.7W
P2(88.25s)	360μJ	400μJ	2500Hz	1.9W

Table.20:Data for switching losses

Through formula(30) and energy calculating formula(31), The final total energy losses through 6 MOSFETs in

single electrical cycle can be measured shown in table.21

$$P_{\text{loss_total}} = P_{\text{loss_sw}} + P_{\text{loss_con}} \quad (30)$$

$$T = \frac{1}{f} \text{ and } P_{\text{total}} = 6 \times \frac{E_{\text{total}}}{T} \quad (31)$$

Operating points	$P_{\text{loss_total}}$	t	E_{total}
P1(3.49s)	44.9W	0.04s	10.776J
P2(88.25s)	10.71W	0.08s	5.1405J

Table.21:Data for total losses

7. Part.G Heat sink

When selecting heat sinks, power loss and air flow rate are crucial considerations because they will directly influence the effectiveness of heat dissipation. in this report the room temperature was setting at 25 degrees Celsius following the data-sheet of the MOSFET_7 which providing a baseline ambient temperature for the heat sink system. The working data for MOSFET_7 shown below in table.22

Table 2 Maximum ratings					
Parameter	Symbol	Values			Unit Note / Test Condition
		Min.	Typ.	Max.	
Continuous DC drain current ⁽¹⁾	I_{DSC}	-	-	238 167	A $T_C = 25^\circ\text{C}$ $T_C = 100^\circ\text{C}$
Peak drain current ⁽²⁾	I_{DM}	-	-	897	A $T_C = 25^\circ\text{C}$, $V_{\text{GS}} = 18\text{ V}$
Avalanche energy, single pulse	E_{AS}	-	-	848	mJ $I_D = 31.8\text{ A}$, $V_{\text{DS}} = 50\text{ V}$; see table 11
Avalanche energy, repetitive	E_{AR}	-	-	4.24	mJ $I_D = 31.8\text{ A}$, $V_{\text{DS}} = 50\text{ V}$; see table 11
Avalanche current, single pulse	I_{AS}	-	-	31.8	A
MOSFET dv/dt ruggedness	dv/dt	-	-	200	V/ns $V_{\text{GS}} = 0 \dots 400\text{ V}$
Gate source voltage (static) ⁽³⁾	V_{GS}	-7	-	23	V
Gate source voltage (transient)	V_{GS}	-10	-	25	V $t_p \leq 500\text{ ns}$, duty cycle $\leq 1\%$
Power dissipation	P_{DS}	-	-	789	W $T_C = 25^\circ\text{C}$
Storage temperature	T_{stg}	-55	-	150	$^\circ\text{C}$
Operating junction temperature	T_J	-55	-	175	$^\circ\text{C}$
Mounting torque	-	-	-	-	Ncm
Continuous reverse drain current ⁽¹⁾	I_{SDC}	-	-	238 155	A $V_{\text{GS}} = 18\text{ V}$, $T_C = 25^\circ\text{C}$ $V_{\text{GS}} = 0\text{ V}$, $T_C = 25^\circ\text{C}$
Peak reverse drain current ⁽²⁾	I_{SM}	-	-	897 276	A $T_C = 25^\circ\text{C}$, $t_p \leq 250\text{ ns}$ $T_C = 25^\circ\text{C}$
Insulation withstand voltage	V_{ISO}	-	-	n.a.	V V_{rms} , $T_C = 25^\circ\text{C}$, $t = 1\text{ min}$

Table 3 Thermal characteristics					
Parameter	Symbol	Values			Unit Note / Test Condition
		Min.	Typ.	Max.	
Thermal resistance, junction - case	$R_{\text{th(j-c)}}$	-	-	0.19	$^\circ\text{C/W}$ Not subject to production test. Parameter verified by design/characterization according to JESD51-14.

Table.22:Working data for MOSFET

$$R_{\text{th_total}} = \frac{T_J - T_{\text{room}}}{P_{\text{loss_total}}} \quad (32)$$

$$R_{\text{th_add}} = R_{\text{th_total}} - R_{\text{th(j-c)}} \quad (33)$$

In the process of selecting heat sinks, an important step is to calculate the additional thermal resistance, First, from the table.22, the thermal resistance $R_{\text{th(j-c)}}$ is 0.19°C/W , and through the formula(32) and(33) the total and additional thermal resistance can be calculated which shown in table.23

Operating points	$R_{\text{th_total}}$	$R_{\text{th_add}}$
P1(3.49s)	3.34°C/W	3.15°C/W
P2(88.25s)	14°C/W	13.81°C/W

Table.23: Data of thermal resistance

Based on the calculated thermal resistance, Heatsink1 was selected as the heat sink for this system, the performance curves of Heatsink1 was shown below in the Figure.28

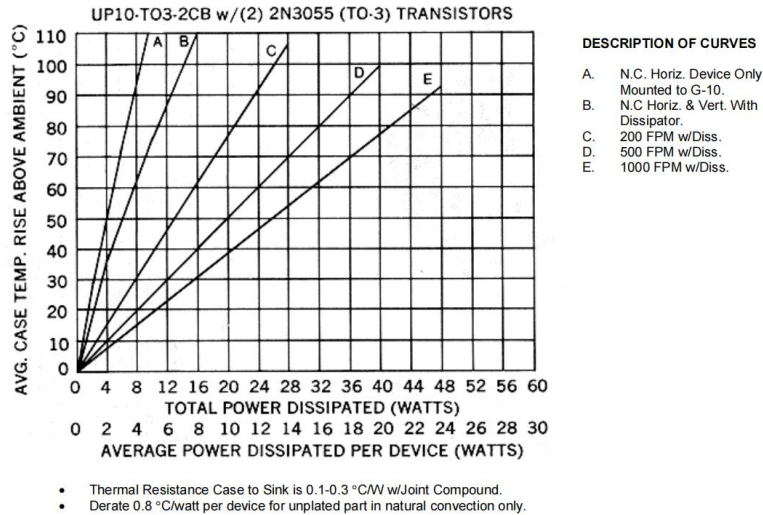


Figure.28: performance curves of Heatsink1

When the maximum power losses reaches 45W at point P1, this heat sink can still maintain the normal operation of the MOSFET. However, this Heatsink1's performance depends on environmental conditions the additional cooling systems, such as fans or air conditioning, can further enhance the system's operational efficiency.

8. Part.H Summary for all elevator system

The elevator system designed in this project is a model of robust, durable, and efficient engineering. The key components such as the motor for vertical movement and the control system for operational management are comprehensive. However, there also have some places that require improvement. Regarding the control system, even though it is fully featured, there may be response delays during the end of acceleration or deceleration phases. This may due to the limitations of 2D simulation, as many real-world factors are not considered in the simulations.

- For instance, the elevator load is almost impossible to maintain consistent torque in practice. To accurately simulate weight changes, it would be necessary to model variations with random arrays, however this feature not implemented in this report.
- In the machine section of this report, after integrating a three-phase motor, the torque and phase voltage and current during deceleration become a non-linear curve with curvature, this may due to an incomplete simulation of the deceleration environment.
- The main challenge of this report is the part of inverter, where a larger number of electrical unit need to consider for how they will affect the harmonics. The data from these electrical components need to be continually tested and paired until finding the data which is closest to the ideal state.